

Part I

## Recitation Questions

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### Problem 1

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February 21, 2026

Problem 1

**Problem 1** Fill in the following blanks with the correct choice of the words from this list:

*Increasing, decreasing, positive, negative, concave up, concave down*

- (a) If you know  $f''(x) > 0$ , then you know  $f'(x)$  is \_\_\_\_\_ and  $f(x)$  is \_\_\_\_\_.

**Explanation.** increasing, concave up

- (b) If you know  $g'(x) < 0$  and decreasing, then you know  $g(x)$  is \_\_\_\_\_ and \_\_\_\_\_.

**Explanation.** decreasing, concave down

- (c) If you know  $h(x)$  is positive, increasing, and concave down, then you know  $h'(x)$  is \_\_\_\_\_ and \_\_\_\_\_ and that  $h''(x)$  is \_\_\_\_\_.

**Explanation.** positive, decreasing, negative

[4]

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## Problem 2

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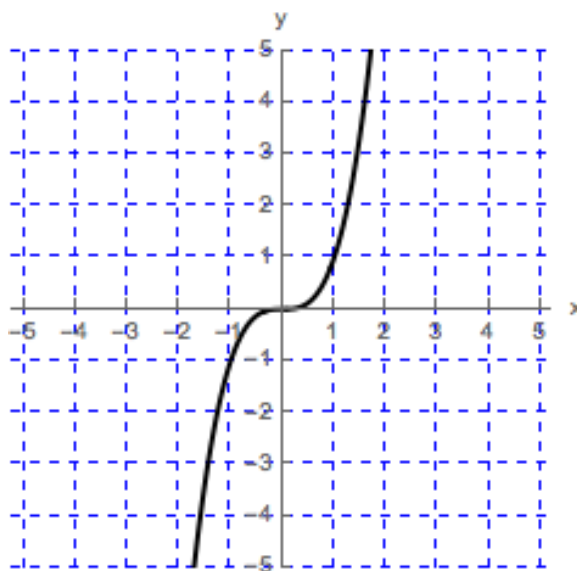
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Problem 2

**Problem 2** Sketch a graph of a function that is continuous on  $(-\infty, \infty)$  that has the following properties.

- (a) Function  $f$  does not have a local maximum or minimum.  $f$  contains a point where  $f'(x) = 0$

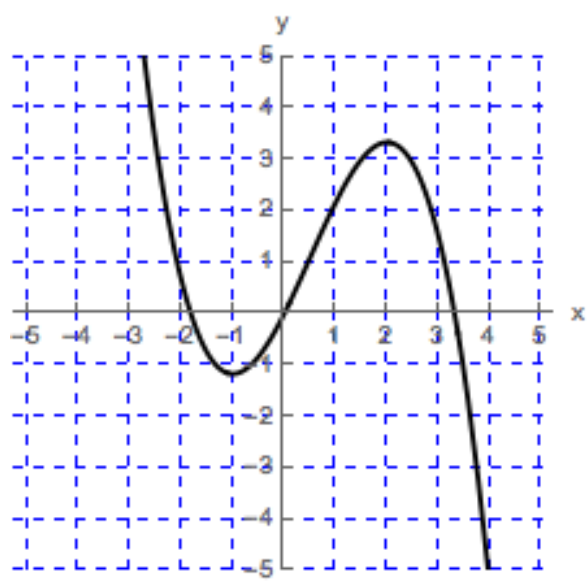
**Explanation.**



- (b)  $g'(x) < 0$  on  $(-\infty, -1)$ ;  $g'(x) > 0$  on  $(-1, 2)$ ;  $g'(x) < 0$  on  $(2, \infty)$ .

**Explanation.**

Problem 2



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# Problem 3

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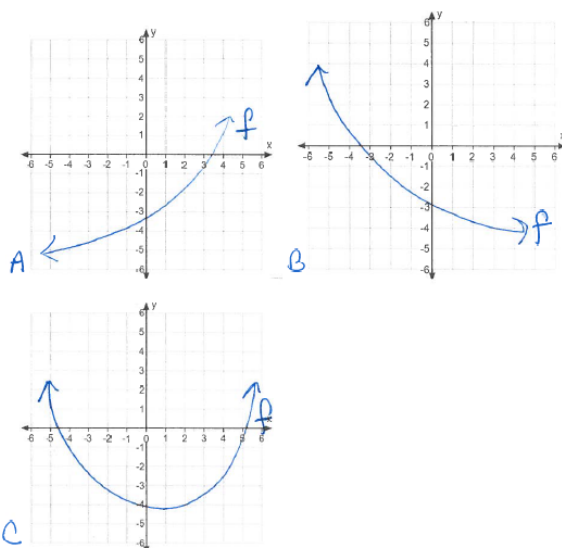
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Problem 3

**Problem 3** Give an example or sketch of a function that is continuous on  $(-\infty, \infty)$  and satisfies given properties. If such a function does not exist, explain why.

- (a) A function  $f$  is concave up and negative everywhere.

**Explanation.** This is not possible. If a function is always concave up then at some point, the function must cross the  $x$ -axis and become positive. See the three example figures below.



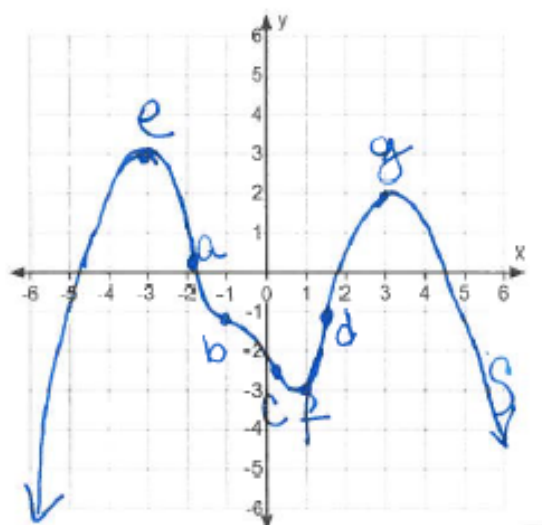
- (b) A function  $f$  is decreasing and concave up everywhere.

**Explanation.** This is possible. See figure B above.

- (c) A function  $s$  has exactly 3 local extrema and four inflection points.

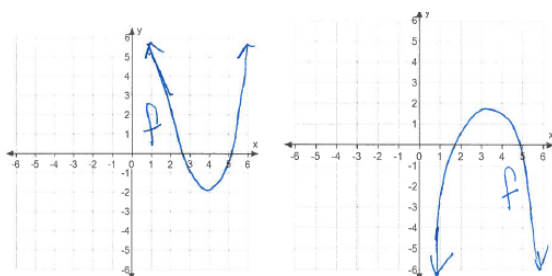
**Explanation.** This is possible. See the figure below. The graph of  $s$  has inflection points at  $a, b, c, d$  and local extrema at  $e, f, g$ .

Problem 3



(d) A function  $f$  has exactly 2 zeros and one local extrema.

**Explanation.** This is possible. See the figures below for examples.





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# Problem 4

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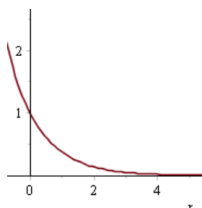
**Problem 4**

(a) You are given that  $f''(x) > 0$  for all  $x$ . Which of the following must be true about  $f(x)$  on the region  $0 \leq x \leq 2$ ?

- (i) There is a critical point between 0 and 2.
- (ii) There is a local maximum, but not enough information is given to
- (iii)  $f$

**Expla**

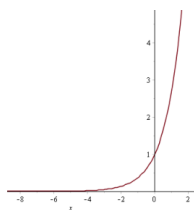
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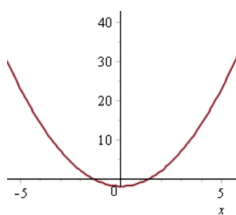
(b) You are told that  $f''(x) > 0$  for all  $x$ . Which of the following must be true about the graph of  $y = f(x)$ ?

- (i) The graph is a straight line.
- (ii) The graph crosses the  $x$ -axis at most once.
- (iii) The graph is concave down.
- (iv) The graph crosses the  $y$ -axis more than once.
- (v) The graph is concave up.

**Explanation.** Only (v) must be true. For (i), the function  $f(x) = e^x$  provides a counterexample:



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Part (iii) is clearly false since  $f''(x) > 0$  means that  $f$  is concave up. Part (iv) is false for any function.

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# Problem 5

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**Problem 5** Suppose a function  $f$  satisfies the following conditions:

- (a)  $f(0) = 0$  and  $f'(-4) = f'(2) = f'(10) = 0$
- (b)  $\lim_{x \rightarrow 6} f(x) = -\infty$ , and  $\lim_{x \rightarrow +\infty} f(x) = 6$
- (c)  $f'(x) < 0$  on  $(-\infty, -4)$ ,  $(2, 6)$ , and  $(10, +\infty)$
- (d)  $f'(x) > 0$  on  $(-4, 2)$ , and  $(6, 10)$
- (e)  $f''(x) > 0$  on  $(-\infty, 0)$ , and  $(14, +\infty)$
- (f)  $f''(x) < 0$  on  $(0, 6)$ , and  $(6, 14)$

- (a) List the **interval(s)** where the function  $f$  is **both increasing and concave UP**.

**Explanation.** ( $-4, 0$ )

- (b) List the **interval(s)** where the function  $f$  is **both increasing and concave DOWN**.

**Explanation.** ( $0, 2$ ),  $(6, 10)$

- (c) List the **interval(s)** where the function  $f$  is **both decreasing and concave UP**.

**Explanation.** ( $-\infty, -4$ ),  $(14, +\infty)$

- (d) List the **interval(s)** where the function  $f$  is **both decreasing and concave DOWN**.

**Explanation.** ( $2, 6$ ),  $(10, 14)$

Problem 5

- (e) List the **x-coordinates** at which  $f$  has a **local minimum**. Write "none" if appropriate.

**Explanation.**  $x = -4$

- (f) List the **x-coordinates** at which  $f$  has a **local maximum**. Write "none" if appropriate.

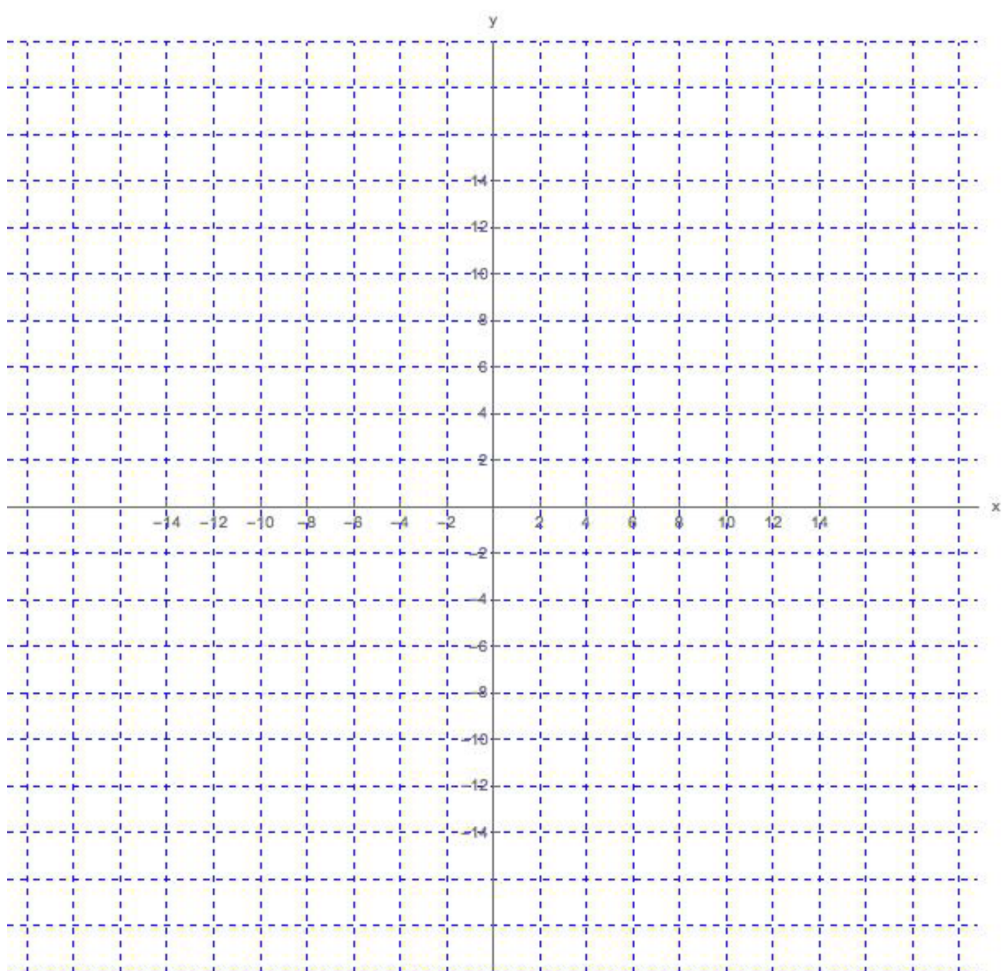
**Explanation.**  $x = 2, \quad x = 10$   
( $f$  can also have a local maximum at  $x = 6$ , if we include  $x = 6$  in the domain)

- (g) List the **x-coordinates** of all **inflection points** of  $f$ . Write "none" if appropriate.

**Explanation.**  $x = 0, \quad x = 14$

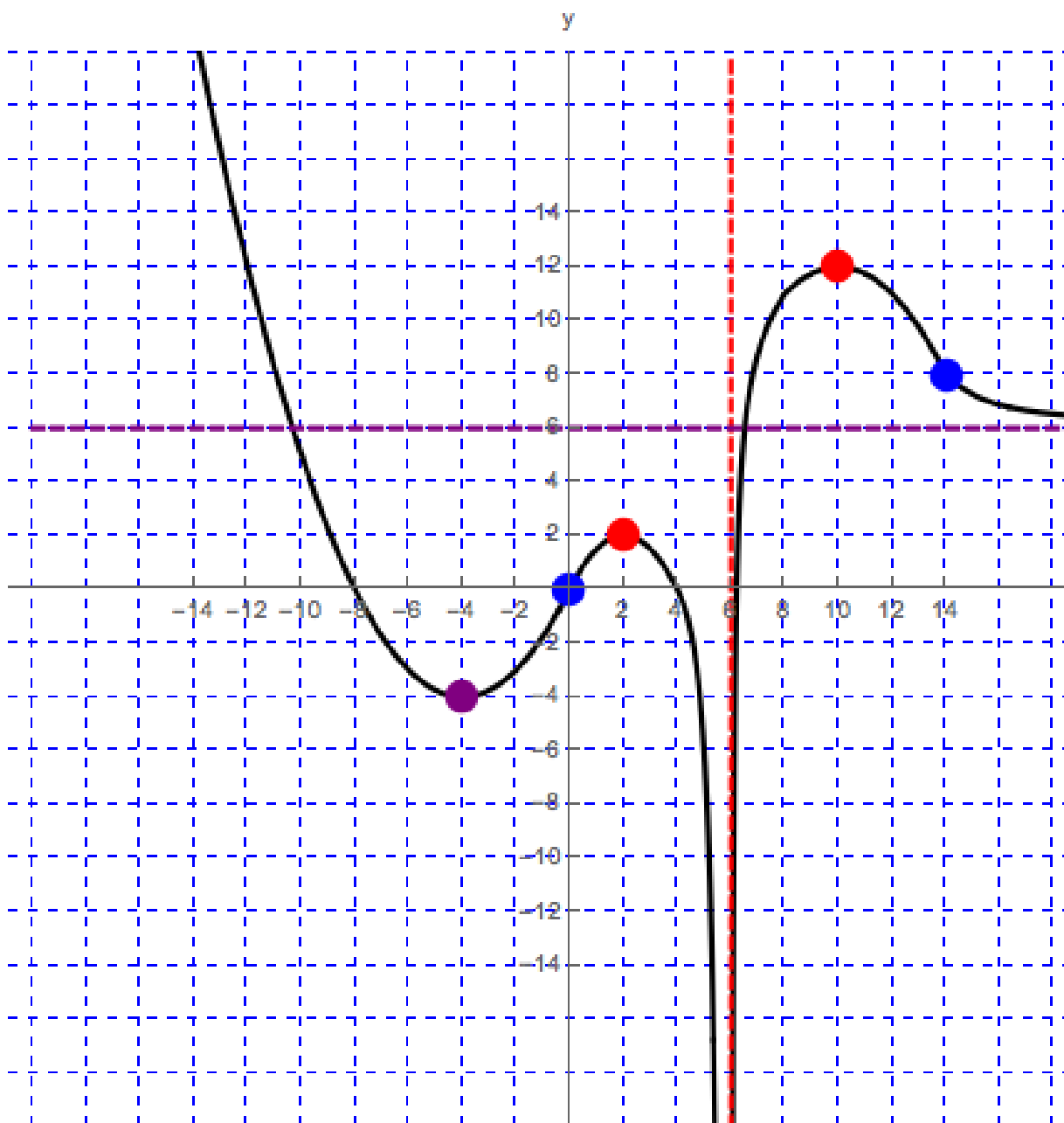
- (h) Sketch the graph of  $f$ .

Problem 5



**Explanation.** This is just one of many possible graphs. The domain of  $f$  could be  $(-\infty, +\infty)$ , and  $f$  could satisfy the given conditions.

Problem 5



Purple point : local minimum

Red points : local maxima



*Problem 5*

Blue points : inflection points

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# Problem 6

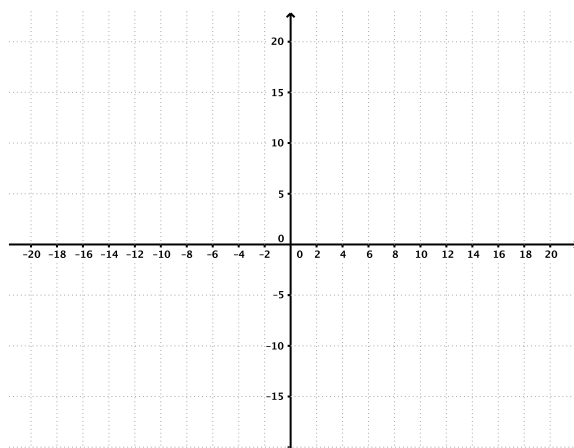
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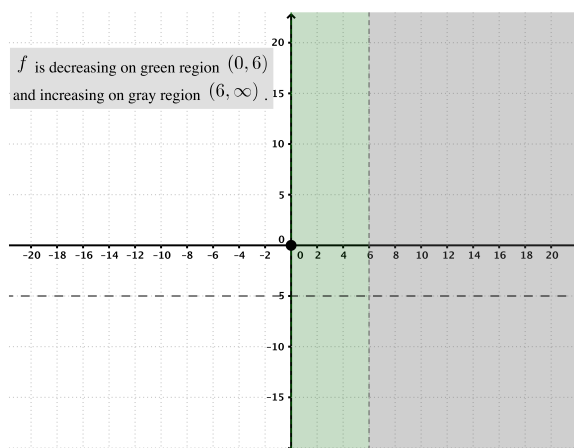
Problem 6

**Problem 6** Sketch the graph of a function  $f$  satisfying all of the conditions:

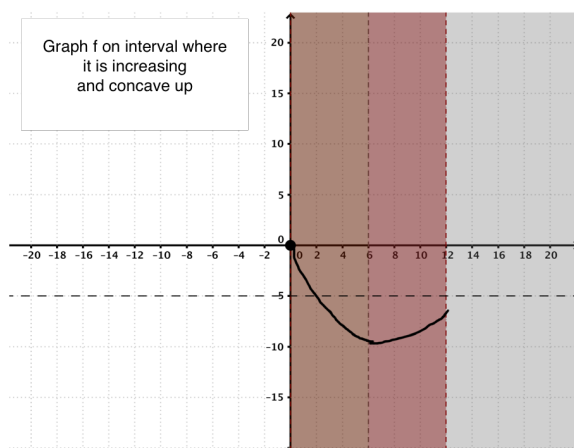
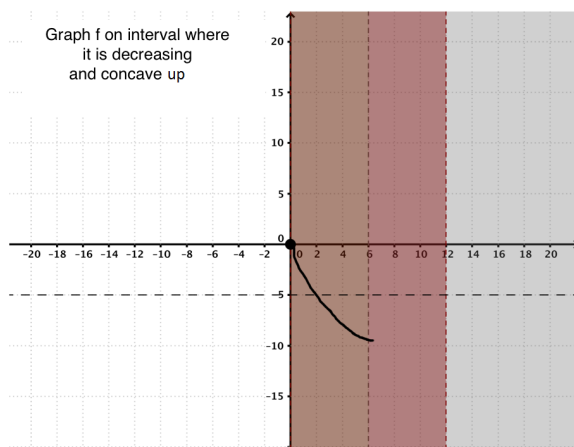
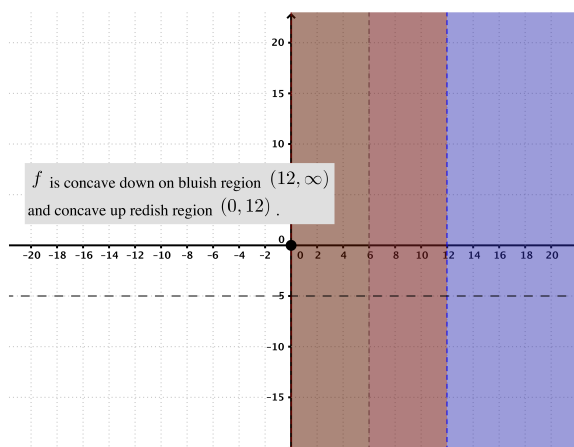
- (a)  $f$  is continuous and odd,  $f(0) = 0$ ,
- (b)  $\lim_{x \rightarrow \infty} f(x) = -5$ ,
- (c)  $f'(x) > 0$  on  $(6, \infty)$ ,
- (d)  $f'(x) < 0$  on  $(0, 6)$ ,
- (e)  $f''(x) > 0$  on  $(0, 12)$ , and
- (f)  $f''(x) < 0$  on  $(12, \infty)$ .



**Explanation.**



Problem 6



Problem 6

